A Quantitative Norm Converse for Co-commutative Quantum Groups

Candidate partial result for arXiv:1709.04873

11 August 2026

Status. Candidate partial result, likely valid. This note answers the source question for all co-commutative locally compact quantum groups, but not for arbitrary locally compact quantum groups.

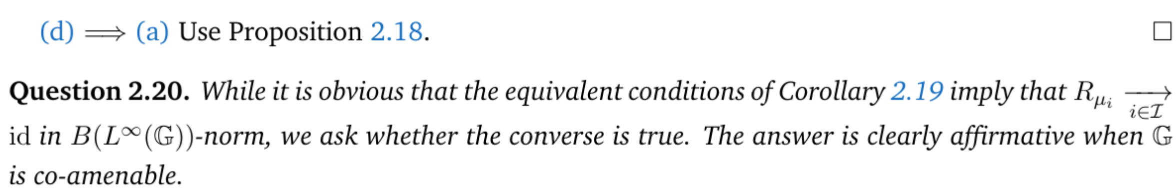
1 Source question

Skalski and Viselter ask in Question 2.20 of [1], PDF page 31, whether the following converse is true. If (μ_i) is a net of states on $C_0^u(\mathbb{G})$ and

$$R_{\mu_i} \longrightarrow \mathrm{id} \quad \text{in } B(L^\infty(\mathbb{G}))\text{-norm},$$

must $\mu_i \to \epsilon$ in $C_0^u(\mathbb{G})^*$-norm? Here ϵ is the universal counit. The source observes that the answer is affirmative when $\mathbb{G}$ is co-amenable.

Figure 1 is a crop of the original source page.



(d) $\Longrightarrow$ (a) Use Proposition 2.18. □

Question 2.20. *While it is obvious that the equivalent conditions of Corollary 2.19 imply that* $R_{\mu_i} \xrightarrow[i\in\mathcal{I}]{} \mathrm{id}$ *in* $B(L^\infty(\mathbb{G}))$*-norm, we ask whether the converse is true. The answer is clearly affirmative when* $\mathbb{G}$ *is co-amenable.*

Figure 1: Question 2.20 in [1, p. 31].

2 Co-commutative specialization

Let H be a locally compact group and put $\mathbb{G} = \widehat{H}$. Then

$$C_0^u(\widehat{H}) = C^*(H), \qquad L^\infty(\widehat{H}) = VN(H).$$

Write $u_s \in M(C^*(H))$ for the canonical universal unitary and $\lambda_s \in VN(H)$ for the left-regular unitary associated with $s \in H$. If μ is a state on $C^*(H)$, its strictly continuous extension to the multiplier algebra defines the normalized continuous positive-definite function

$$\varphi(s) = \mu(u_s).$$

Since the coproduct is group-like on the canonical unitaries, the reduced right convolution operator satisfies

$$R_\mu(\lambda_s) = \varphi(s)\lambda_s \qquad (s \in H). \tag{1}$$

3 Idea of the proof

Equation (1) converts norm closeness of R_μ to the identity into uniform closeness of every positive-definite coefficient $\varphi(s)$ to 1. In the GNS representation, the orbit of the cyclic vector therefore lies in a thin spherical cap. The unique minimum-norm point in the closed convex hull of that orbit is fixed by the whole group. More precisely, it is the orthogonal projection of the cyclic vector onto the invariant subspace, and the spherical-cap estimate forces this projection to have squared norm at least $1-\delta$.

Normalizing the projection realizes the counit as a vector state inside the same representation as μ. The elementary distance formula for two vector states then gives a $2\sqrt{\delta}$ bound. No averaging measure or invariant mean is used, so the argument remains valid when H is nonamenable.

4 Partial theorem and proof

Theorem 1. *Let H be a locally compact group. For every state μ on $C^*(H)$, let R_μ be its reduced right convolution operator on $VN(H)$ and let ϵ be the counit, equivalently the state induced by the trivial representation of H. Then*

$$\|\mu-\epsilon\| \le 2\sqrt{\|R_\mu - \mathrm{id}\|_{B(VN(H))}}. \tag{2}$$

Proof. Set

$$\delta = \|R_\mu - \mathrm{id}\|_{B(VN(H))}.$$

By (1) and $\|\lambda_s\| = 1$,

$$|\varphi(s)-1| = \|(R_\mu - \mathrm{id})(\lambda_s)\| \le \delta \qquad (s \in H). \tag{3}$$

Let (π, K, ξ) be the GNS representation of μ, with $\|\xi\| = 1$, so that $\varphi(s) = \langle \pi(u_s)\xi, \xi\rangle$. Let $\mathcal{C}$ be the closed convex hull of the orbit $\{\pi(u_s)\xi : s \in H\}$. The Hilbert-space norm has a unique minimum on the nonempty closed convex set $\mathcal{C}$; denote its minimizer by η. Each $\pi(u_t)$ maps $\mathcal{C}$ onto itself and preserves the norm, so uniqueness gives

$$\pi(u_t)\eta = \eta \qquad (t \in H).$$

Thus η is invariant.

Let K^H be the invariant subspace and let P be the orthogonal projection onto K^H. For every $v \in K^H$, every orbit vector has the same inner product with v as ξ does. By convexity and closure,

$$\langle \eta, v\rangle = \langle \xi, v\rangle \qquad (v \in K^H).$$

Since $\eta \in K^H$, this identity says exactly that $\eta = P\xi$.

From (3), every orbit vector $w = \pi(u_s)\xi$ satisfies $\mathrm{Re}\langle w, \xi\rangle \ge 1-\delta$. The same inequality holds throughout $\mathcal{C}$, hence at $P\xi$. Because P is an orthogonal projection,

$$\|P\xi\|^2 = \langle P\xi, \xi\rangle \ge 1-\delta. \tag{4}$$

Suppose first that $\delta < 1$. Then $P\xi \ne 0$, and

$$\zeta = \frac{P\xi}{\|P\xi\|}$$

is a unit invariant vector. Its vector state on $C^*(H)$ is the state of the trivial representation, hence it is ϵ. Thus μ and ϵ are the restrictions to $\pi(C^*(H))$ of the vector states of ξ and ζ. The distance of the restrictions is bounded by the distance of the two vector states on $B(K)$. The latter is the trace norm of the difference of the two rank-one projections and equals

$$2\sqrt{1-|\langle \xi,\zeta\rangle|^2}.$$

Since P is an orthogonal projection, $|\langle \xi,\zeta\rangle| = \|P\xi\|$. Therefore (4) gives

$$\|\mu-\epsilon\| \le 2\sqrt{1-\|P\xi\|^2} \le 2\sqrt{\delta}.$$

If $\delta \ge 1$, then (2) follows from the elementary bound $\|\mu-\epsilon\| \le 2 \le 2\sqrt{\delta}$ for two states. This proves the theorem in all cases. □

Corollary 2. *For every locally compact group H and every net (μ_i) of states on $C^*(H)$,*

$$R_{\mu_i} \longrightarrow \mathrm{id} \quad \textit{in } B(VN(H))\textit{-norm} \quad \Longrightarrow \quad \mu_i \longrightarrow \epsilon \quad \textit{in } C^*(H)^*\textit{-norm.}$$

Consequently Question 2.20 has an affirmative answer for every co-commutative locally compact quantum group $\widehat{H}$.

Proof. Apply Theorem 1 to each μ_i and pass to the limit. □

5 Verification notes

- The proof uses no amenability hypothesis. In particular, if H is nonamenable, then $\widehat{H}$ is not co-amenable, so the corollary goes beyond the case recorded in the source paper.
- The closed-convex-hull argument does not invoke a Følner net or an invariant mean. Invariance follows solely from uniqueness of the minimum-norm point.
- The identity $\eta = P\xi$ is not assumed: it follows because all orbit vectors, and therefore every point of $\mathcal{C}$, have the same inner products with invariant vectors.
- The vector-state estimate is applied first on $B(K)$ and then restricted to $\pi(C^*(H))$, so no purity assertion about the states on $C^*(H)$ is needed.
- No numerical or symbolic computation is used.

Verifier verdict: likely valid. The two most useful checks are the multiplier formula (1) under the source's right convolution convention and the rank-one vector-state calculation in the last paragraph of the proof.

6 Scope and novelty status

The argument depends crucially on the group-like unitaries λ_s and does not provide an analogous orbit for a general locally compact quantum group. The full Question 2.20 therefore remains open in this packet.

On 11 August 2026, the local lightweight indexes were searched for the exact arXiv id, the question, and its core terms. The locally parsed arXiv corpus was searched for later citations to arXiv:1709.04873. Bounded official-arXiv searches used the exact id, title, and combinations of "right convolution", "counit", "operator norm", "co-amenable", and "co-commutative". No later paper explicitly answering Question 2.20 and no statement of the quantitative co-commutative subcase was found. Novelty confidence is moderate, because the argument is elementary after specialization to group duals.

7 Human review recommendation

Check the two points named in the verifier verdict. If accepted, retain this as a substantial partial answer covering the whole co-commutative class, including non-co-amenable examples. It must not be described as settling the general locally compact quantum-group question.

References

[1] Adam Skalski and Ami Viselter, *Convolution semigroups on locally compact quantum groups and noncommutative Dirichlet forms*, arXiv:1709.04873, 2017. `https://arxiv.org/abs/1709.04873`